

The Magnetic Field

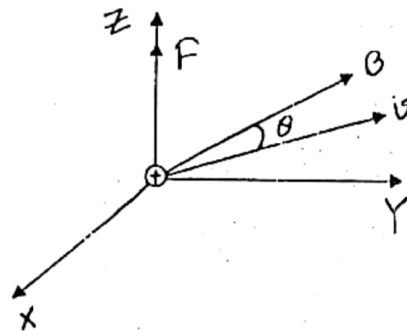
The Magnetic Field:

The Space around the current carrying conductor is defined as the site of a magnetic field. Magnetic field is a vector field whose magnitude and direction at any point specified by a vector $\vec{B}$ called magnetic induction.

The Definition of B:

Let us consider a charge q moving with velocity $\vec{v}$ in a region where the magnetic induction, $\vec{B}$ exists (Fig.-01). The charged particle experiences a force $\vec{F}$ in a direction perpendicular to both $\vec{v}$ and $\vec{B}$. Therefore, we can write,

$$\vec{F} = q(\vec{v} \times \vec{B}) \dots \dots \dots (i)$$



The magnitude of the magnetic deflecting force, F according to the rules for vector products is given by,

$$F = qBv \sin\theta$$

$$\Rightarrow B = \frac{F}{q(v \sin\theta)} \dots \dots \dots (ii)$$

So, the magnitude of magnetic induction, B is

$$B = \frac{\text{Force on the charged particle}}{\text{Charge} \times \text{component of the velocity perpendicular to } \vec{B}}$$

The S.I. unit of $\vec{B}$ is called tesla (T).

Miracle Point
জাহাঙ্গীর সেন্টার (সীচ ভগা)
অধ্যাপক সাহান, মল্লিক রোড, চকরাবাহা, চকরা।
মধ্যকারি, কম্পিউটার কলেজ, ক্রিষ্টি
মহাস্থান আরোপন এড ব্রেনারি
সংগ্রহ নং-01558276636

Magnetic Flux:

The magnetic flux ϕ through a surface S is defined as

$$\phi = \int_S B \cdot dS$$

It represents the lines of induction crossing surface S .

The S.I. unit of magnetic flux is a weber (Wb).

If B is uniform and normal to area A ,

$$\phi = BA$$

If $A = 1\text{m}^2$, $\phi = B$. i.e., magnetic induction is numerically to normal flux per unit area. Therefore, it is also called magnetic flux density.

Lorentz Force on a Moving Charge:

Suppose a charge q moves with a velocity v through a region where both electric field E and magnetic field B are present. Then the resultant force F on the moving charge is

$$F = qE + q(v \times B) = q(E + v \times B)$$

This equation is called Lorentz force equation.

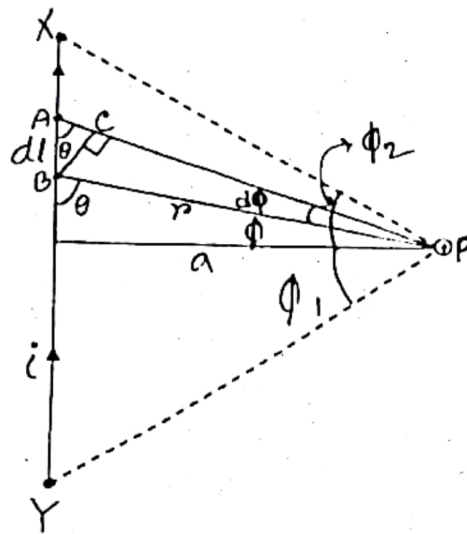
Magnetic Induction at a point due to a Straight Conductor Carrying Current:

Let us consider a straight conductor XY carrying a current i in the direction Y to X (Fig.-01). P is a point at a perpendicular distance a from the conductor. Consider an element AB of length dl . Let $BP=r$ and $\angle OBP = \theta$.

Magnetic induction at P due to the element AB

$$= dB$$

$$= \left(\frac{\mu_0}{4\pi} \right) \frac{i dl \sin\theta}{r^2} \dots \dots \dots (i)$$



From B, draw BC perpendicular to PA.

Let $\angle OPB = \phi$, $\angle BPA = d\phi$.

Then, $BC = dl \sin\theta = r d\phi$

Therefore, we can write

$$dB = \left(\frac{\mu_0}{4\pi} \right) \frac{i r d\phi}{r^2}$$

$$= \left(\frac{\mu_0}{4\pi} \right) \frac{i d\phi}{r}$$

In ΔOPB , $\cos\phi = \frac{a}{r}$

$$\Rightarrow r = \frac{a}{\cos\phi}$$

$$\therefore dB = \left(\frac{\mu_0}{4\pi}\right) \frac{i \cos\phi d\phi}{a} \dots \dots \dots (ii)$$

The direction of dB will be perpendicular to the plane containing dl and r . It will be directed into the page at P as shown by right hand rule.

Let ϕ_1 and ϕ_2 be the angles made by the ends of the wire at P . Then, magnetic induction at P due to the whole conductor is

$$\begin{aligned} B &= \int_{-\phi_1}^{\phi_2} \left(\frac{\mu_0}{4\pi}\right) \frac{i \cos\phi d\phi}{a} \\ &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi]_{-\phi_1}^{\phi_2} \\ &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi_2 - \sin(-\phi_1)] \\ &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi_2 + \sin\phi_1] \dots \dots \dots (iii) \end{aligned}$$

If the conductor is infinitely long, $\phi_1 = \phi_2 = 90^\circ$.

$$\begin{aligned} \therefore B &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [1 + 1] \\ \Rightarrow B &= \frac{\mu_0 i}{2\pi a} \dots \dots \dots (iv) \end{aligned}$$

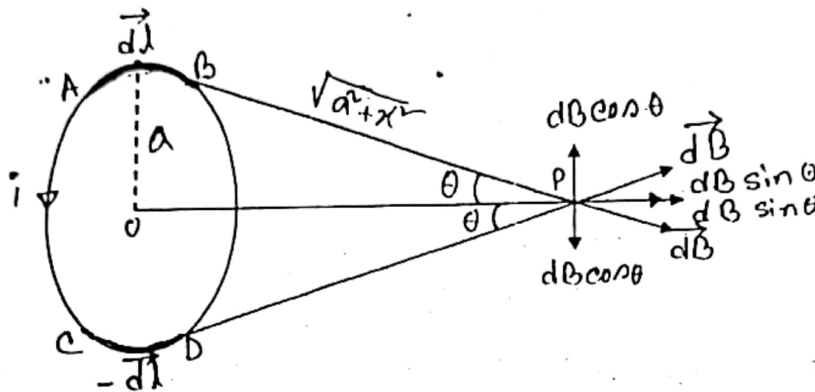
Magnitude of B depends on i and a .

$$\therefore B \propto \frac{1}{a}$$

The lines of B form concentric circles around the wire (Fig.-02).

Magnetic Induction at a point on the Axis of a Circular Coil Carrying Current:

Let us consider a circular coil of radius a , carrying a current i (Fig.-01). P is a point on its axis at a distance x from the centre O . Consider two opposite current elements AB and CD each of length dl . The distance of P from any point on the circumference of the coil is $\sqrt{a^2 + x^2}$.



The field at P due to $AB = dB = \left(\frac{\mu_0}{4\pi} \right) \frac{i dl \sin 90^\circ}{(a^2 + x^2)}$

$$= \left(\frac{\mu_0}{4\pi} \right) \frac{i dl}{(a^2 + x^2)} \dots \dots \dots (i)$$

($\because$ the direction of the current is at right angles to the line joining P to AB)

This is in the direction PL , perpendicular to the line joining the midpoint of AB with P . Considering the element CD , the magnitude of dB at P due to this element is the same as that given in equation (i). But, it is directed along PM . E is the midpoint of AB . Let $\angle EPO = \theta$.

The component $dB \cos \theta$ perpendiculars to the axis of the coil and due to the two opposite elements cancel each other. But components $dB \sin \theta$

along the axis are in the same direction. Thus, the total magnetic induction at P due to the entire coil

$$\begin{aligned}
 B &= \int_{l=0}^{l=2\pi a} dB \sin\theta \\
 &= \int_{l=0}^{l=2\pi a} \left(\frac{\mu_0}{4\pi} \right) \frac{i dl \sin\theta}{(a^2 + x^2)} \\
 &= \int_{l=0}^{l=2\pi a} \left(\frac{\mu_0}{4\pi} \right) \frac{i dl}{(a^2 + x^2)^{3/2}} \times \frac{a}{(a^2 + x^2)^{1/2}} \left[\because \sin\theta = \frac{a}{(a^2 + x^2)^{1/2}} \right] \\
 &= \frac{\mu_0}{4\pi} \frac{ia}{(a^2 + x^2)^{3/2}} \times 2\pi a \\
 \therefore B &= \frac{\mu_0 ia^2}{2(a^2 + x^2)^{3/2}}
 \end{aligned}$$

This is the required magnetic field due to a circular current coil loop.

If the coil has N turns, then

$$\therefore B = \frac{\mu_0 Nia^2}{2(a^2 + x^2)^{3/2}}$$

At the centre of the coil, $x=0$

$$\therefore B = \frac{\mu_0 Ni}{2a}$$

When, $x \gg a$,

$$B = \frac{\mu_0 Nia^2}{2x^3}$$

$$\oint \vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2\pi r} (2\pi r) = \mu_0 i$$

Thus the integral $\oint \vec{B} \cdot d\vec{l}$ is μ_0 times current through the area bounded by the circle. This is Ampere's law.

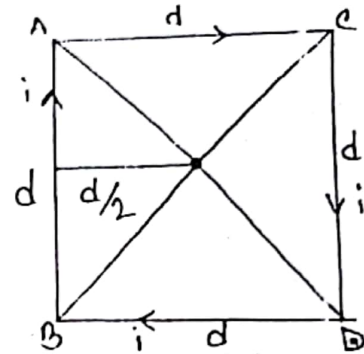
Problem:-1

A square coil of side "d" carries a current "i". Calculate the magnetic induction at the centre of the coil.

Solution: ABCD is a square coil carrying a current i as shown in Fig.-

Magnetic induction at the centre due to AB is

$$\begin{aligned} B' &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} |\sin\phi_2 + \sin\phi_1| \\ &= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{\frac{d}{2}} [\sin\phi_2 + \sin\phi_1] \\ \therefore B' &= \frac{\mu_0 i}{\sqrt{2} \pi d} \end{aligned}$$



The magnetic induction due to current in the remaining three arms of the square loop will also have the same magnitude and direction. Thus, total magnetic induction at the centre due to the square coil is

$$B = 4B' = \frac{2\sqrt{2}}{\pi d} \mu_0 i$$

Problem:-2

Find the magnetic induction at the centre of a square current loop of side 1 metre carrying a current of 1 ampere. Ans: $8\sqrt{2} \times 10^{-7} \text{ wb m}^{-2}$.

Problem:-3

A circular coil has a radius of 0.1m and a number turns of 50. Calculate the magnetic induction at a point

(i) on the axis of the coil and distance 0.2 m from the centre.

Ans: $2.81 \times 10^{-6} \text{ T}$.

(ii) at the centre of the coil, when a current of 0.1 A flows in it.

Ans: $3.14 \times 10^{-5} \text{ T}$.

Faraday's law of electromagnetic Induction:

Statement of Faraday's law:

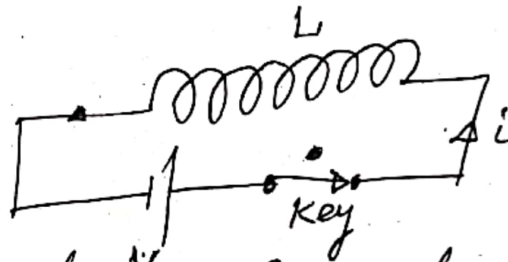
"Faraday's law states that the electromotive force $\mathcal{E}$ induced in a circuit equals the negative of the rate of change of magnetic flux Φ_B through the circuit", which can be written mathematically as—

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

Here, the -ve sign indicates the induced e.m.f. is in the opposite direction of the rate of change of flux.

For N -number of turns, the induced e.m.f. is, $\mathcal{E} = -N \frac{d\Phi_B}{dt}$

Self Inductance:



If N be the number of lines of force linked with the coil at the same instant, it is proportional to the current ' i ' of the circuit i.e., $N \propto i$

$$N = Li$$

where, L is the self inductance of the coil. The self inductance of a coil is equal to the number of lines of force linked with the coil when a unit current flows through it.

When a current 'i' flows through the coil the work done is equal to--

$$dW = \mathcal{E} i dt, \quad \text{where, } \mathcal{E} \text{ is the induced e.m.f.}$$

$$\begin{aligned} dW &= - \frac{d\Phi_B}{dt} \cdot i dt \\ &= - \frac{dN}{dt} \cdot i dt \\ &= - \frac{d}{dt} (Li) i dt \\ &= - L \frac{di}{dt} \cdot i dt \\ &= - Li di \end{aligned}$$

The total work done when the current varies from 0 to i is--

$$\begin{aligned} W &= \int_0^i dW = \int_0^i - Li di \\ &= - \frac{Li^2}{2} \end{aligned}$$

$$\text{If } i = 1, \quad 2W = L$$

The inductance of a coil is twice the amount of work done in establishing a unit current in the circuit.

Mutual Inductance

If two coils A and B are connected side by side or one wound over the other, the current in coil A produces magnetic lines of force and linked with the coil B.
Then

$$N \propto i$$

$$N = M i$$

where, M is the mutual inductance

Def: of M :

The mutual inductance M between two coils is the flux linked with the one when the unit current flows through the other.

Calculation of $\vec{B}$ for a long solenoid by using Ampere's Law:

Solenoid: A wire of tightly wound coil is called a solenoid.

Calculation of $\vec{B}$ for a long solenoid:



A cross-sectional view of a solenoid.

Let we consider a solenoid whose cross-sectional view is shown in Fig. The dot point indicates the incoming current and the cross mark shows the outgoing current through the winding coil. The direction of magnetic field induction $\vec{B}$ is at the right of the solenoid.

Let we consider a rectangular area 'abcd' of length 'l'.

By applying Ampere's Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i \quad \dots (1)$$

Ampere's Law for the same solenoid as:

$$\oint \vec{B} \cdot d\vec{s} = \int_a^b \vec{B} \cdot d\vec{s} + \int_b^c \vec{B} \cdot d\vec{s} + \int_c^d \vec{B} \cdot d\vec{s} + \int_d^a \vec{B} \cdot d\vec{s}$$

we use Ampere's law for each term:

1st term: $\int_a^b \vec{B} \cdot d\vec{s} = \int_a^b B ds \cos 0^\circ = \int_a^b B ds = B \int_a^b ds = Bh$

2nd term: $\int_b^c \vec{B} \cdot d\vec{s} = 0$, because B and ds is at right angles. $B \cdot ds = B ds \cos 90^\circ = 0$

3rd term: $\int_c^d \vec{B} \cdot d\vec{s} = 0$, because B is zero, outside the solenoid.

4th term: $\int_d^a \vec{B} \cdot d\vec{s} = 0$, because B and ds is again at right angles.

only the 1st term will remain and the Ampere's law becomes -

$$\oint \vec{B} \cdot d\vec{s} = \int_a^b \vec{B} \cdot d\vec{s} = Bh = \mu_0 i_{\text{enclosed}}$$

Here, i_{enclosed} = current for the solenoid.
 $= i(nh)$

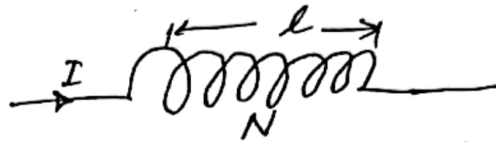
where, n = no. of turns per unit length
 h = length of the solenoid.

$$Bh = \mu_0 i n h$$

$$\therefore B = \frac{\mu_0 i n h}{h} = \mu_0 i n$$

Self Inductance due to a long solenoid.

Let we consider a long air-cored solenoid of length l , total number of turns N and area of cross-section A .
The number of turns per unit length is $\frac{N}{l}$



Let we consider a current I flows through the solenoid and the magnetic field inside the solenoid is

$$\vec{B} = \mu_0 \left(\frac{N}{l} \right) I$$

The flux through each turn is, $\phi_B = BA = \frac{\mu_0 N I A}{l}$

Total flux through N turns,

$$\phi = \phi_B N = \frac{\mu_0 N^2 I A}{l}$$

$\therefore$ Self inductance of the solenoid is -

$$L = \frac{\phi}{I} = \frac{\mu_0 N^2 A}{l}$$

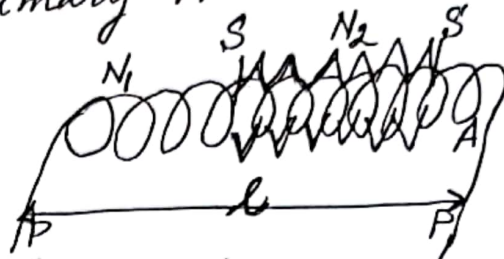
If the solenoid is wound on a core of constant permeability μ , then

$$L = \frac{\mu N^2 A}{l}, \text{ where } \mu = \mu_0 \mu_r$$

μ_r = relative permeability.

Mutual Inductance between two coaxial solenoids.

Let we consider a long air-cored solenoid with primary PP and secondary SS' (shown in fig.)



Let N_1 = no. of turns in the primary.
 N_2 = no. of turns in the secondary.
 A = area of the cross-section
 l = length of the primary coil,
 I = current in the primary coil

magnetic flux field
at any point inside
the primary } $B = \mu_0 \left(\frac{N_1}{l} \right) I$

$\therefore \frac{N_1}{l}$ = no. of turns per unit length.

$\therefore$ magnetic flux through each turn of the primary coil is: $\phi = BA = \mu_0 \left(\frac{N_1}{l} \right) IA$

Since the secondary is wound closely over the central portion of the primary, the same flux is also linked with each turn of the secondary,

magnetic flux through each
turn of the secondary } $= \mu_0 \left(\frac{N_1}{l} \right) IA$

P. T. O.

✓ ~~mutual inductance~~
 Total magnetic flux through N_2 turns of the secondary $= \phi = \mu_0 \left(\frac{N_1}{l} \right) I A N_2$

$\therefore$ The mutual inductance

$$M = \frac{\phi}{I} = \frac{\mu_0 \left(\frac{N_1}{l} \right) I A N_2}{I}$$

$$= \frac{\mu_0 N_1 N_2 A}{l}$$

If the core is a material of permeability μ_r , then

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{l} \text{ henry.}$$

If there are a number of areas of cross-section A_1, A_2, A_3 etc. and relative permeabilities $\mu_{r1}, \mu_{r2}, \mu_{r3}$ etc,

$$M = \frac{\mu_0 N_1 N_2}{l} [\mu_{r1} A_1 + \mu_{r2} A_2 + \mu_{r3} A_3 + \dots] \text{ henry.}$$